Lecture 15.

- Exponentiation gadget:

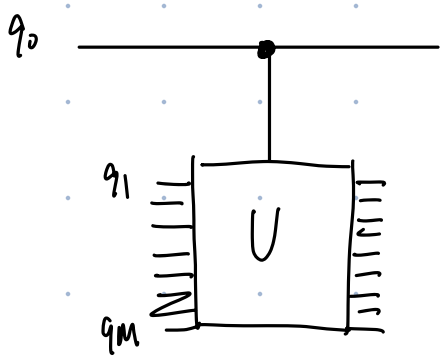
given a controlled unitary, can we exponentiate that unitary?

$e^{-i\frac{\theta}{2}U}$ ← needs to be Hermitian

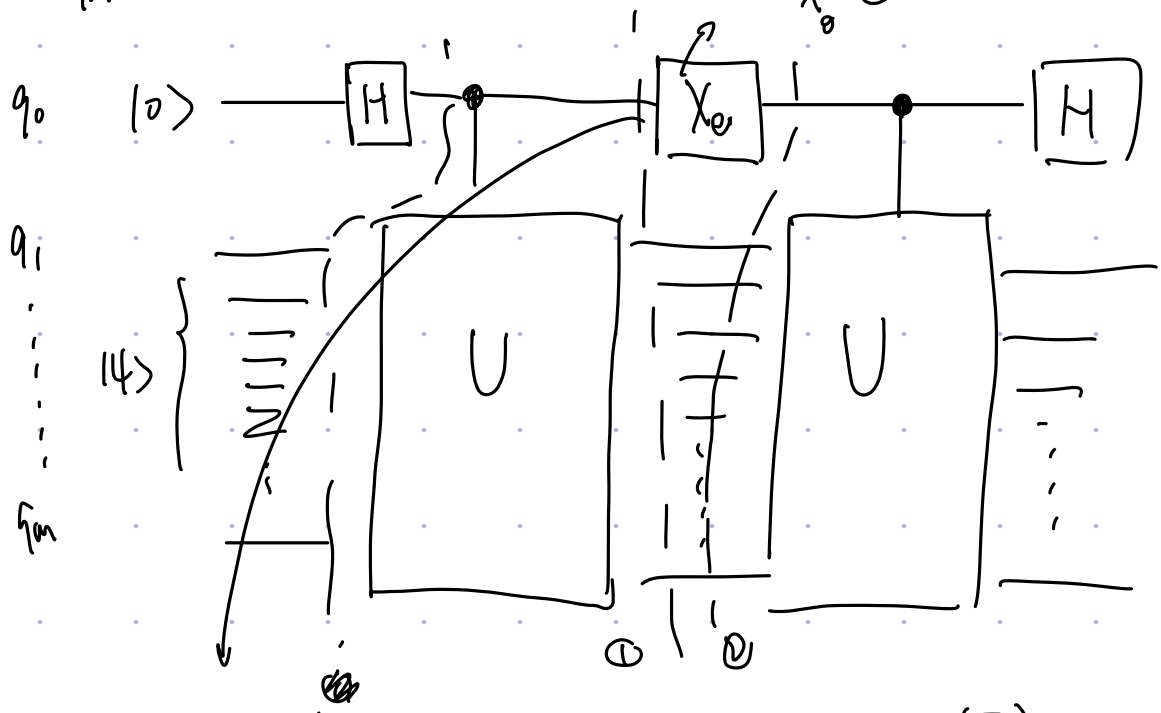
examples: X, Y, Z, $X \otimes Y \otimes Z$

$e^{-i\frac{\theta}{2}\hat{n}\cdot\vec{\sigma}}$

$\underbrace{e^{-i\frac{\theta}{2}X}, Y, Z}_{\text{native gates.}}$



$X_\theta = e^{i\theta X}$

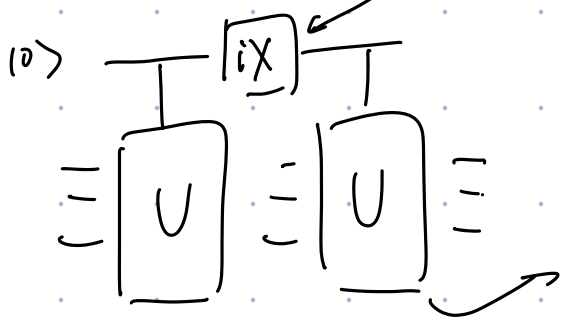
- 

$e^{i\theta U}|\psi\rangle$

The circuit produces a superposition of $\cos\theta I + i\sin\theta U$.

when $\theta = 0$, should get identity (I)

when $\theta = \pi/2$, should get iU



unitary gets applied one but time.

to the circuit

At ①: $\frac{1}{\sqrt{2}}\left[|\psi\rangle \otimes |0\rangle + U|\psi\rangle \otimes |1\rangle \right]$

At ②: $\frac{1}{\sqrt{2}}\Big[|\psi\rangle \otimes \left[\cos\theta|0\rangle + i\sin\theta|1\rangle\right] +$

$U|\psi\rangle \otimes \left[\cos\theta|1\rangle + i\sin\theta|0\rangle\right]\Big]$

$\frac{1}{\sqrt{2}}\Big[\cos\theta|\psi 0\rangle + i\sin\theta|\psi 1\rangle +$

$\cos\theta \cdot U|\psi\rangle \otimes |1\rangle + i\sin\theta\, U|\psi\rangle \otimes |0\rangle\Big]$

At ③: $\frac{1}{\sqrt{2}}\Big[\cos\theta|\psi 0\rangle + i\sin\theta\, U|\psi\rangle \otimes |1\rangle +$

$\cos\theta \cdot U^2|\psi\rangle \otimes |1\rangle + i\sin\theta\, U|\psi\rangle \otimes |0\rangle\Big]$

$= \frac{1}{\sqrt{2}}\Big[(\cos\theta + i\sin\theta U)|\psi\rangle \otimes |0\rangle +$

$(\cos\theta + i\sin\theta U)|\psi\rangle \otimes |1\rangle\Big]$

$= \frac{1}{\sqrt{2}}\Big[(\cos\theta + i\sin\theta U)|\psi\rangle \otimes (|0\rangle + |1\rangle)\Big]$

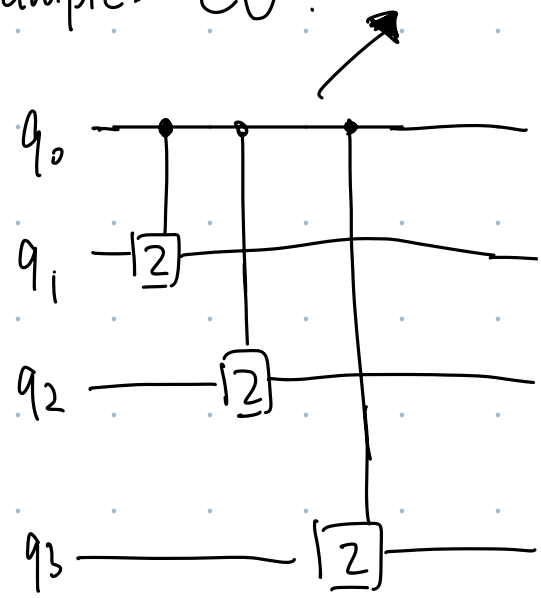
$= e^{i\theta U}|\psi\rangle \otimes |+\rangle.$

At ④: $= e^{i\theta U}|\psi\rangle \otimes H|+\rangle = e^{i\theta U}|\psi\rangle \otimes |0\rangle$

Now I see! (Need algebra to see)

$X_\theta \to$ exponential 2×2 matrix $\Rightarrow U$: exponential $M\times M$ matrix.

Example: cU. How do you actually do 2 qubit gates.



$q_0 = |0\rangle \Rightarrow U = I \otimes I \otimes I$

$q_1 = |1\rangle \Rightarrow U = Z \otimes Z \otimes Z$.

$cU = cZZZ$

$e^{i\theta U} = e^{i\theta Z\otimes Z\otimes Z}$

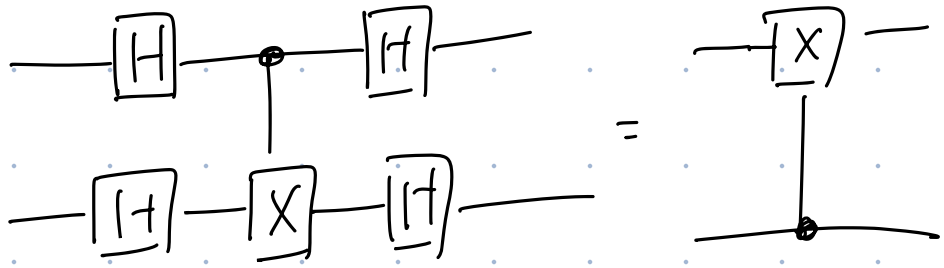
$$U\underbrace{(|+\rangle \otimes |+\rangle \otimes |+\rangle)}_{\text{doesn't do entangling}} = \underbrace{|-\rangle \otimes |-\rangle \otimes |-\rangle}$$

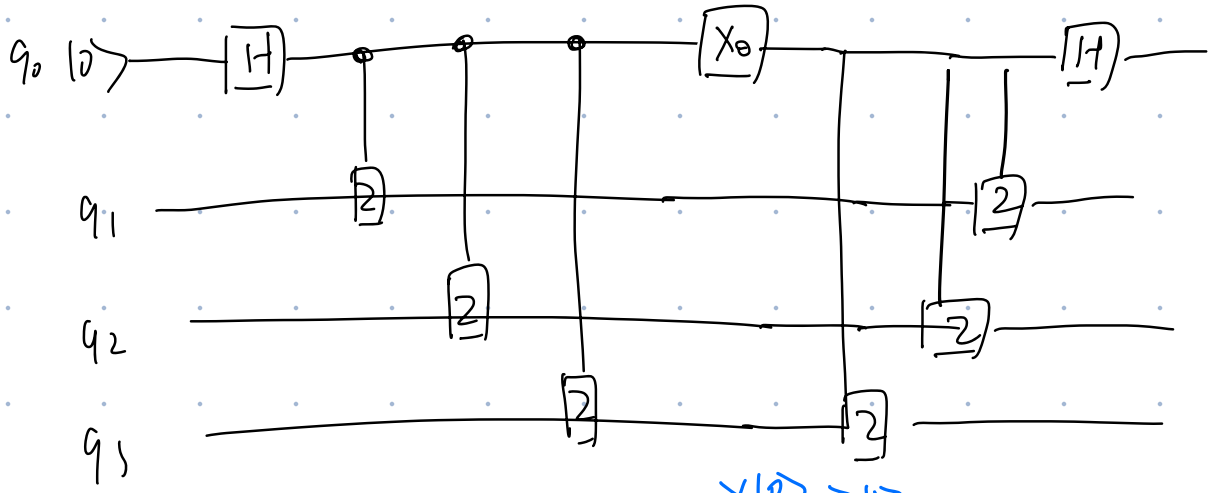
$$ZZZ(\theta) = e^{i\theta Z\otimes Z\otimes Z} \underbrace{= \cos\theta I + i\sin\theta (Z\otimes Z\otimes Z)}_{\text{nontrivial, applies a superposition}}$$

$$ZZZ(\theta)|+++\rangle = \underbrace{\cos\theta |+++\rangle + i\sin\theta |---\rangle}$$

entangled state, unless $\theta = 0, \pi$.

Different version related to the identity.



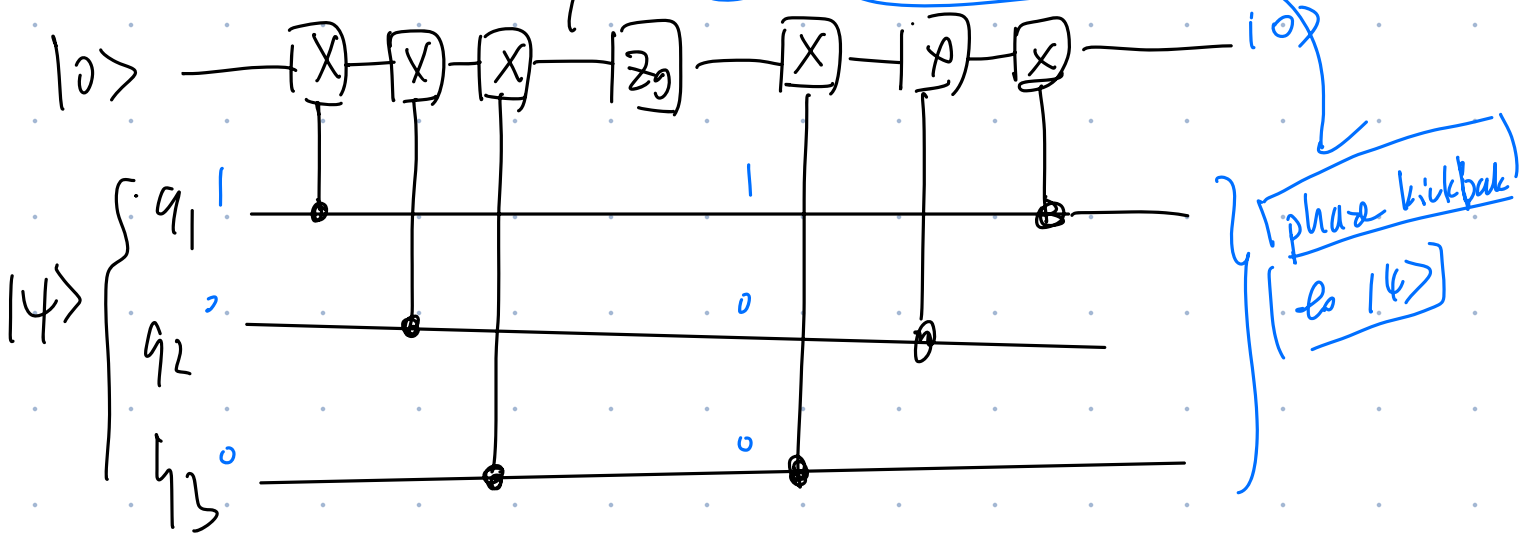


$X|0\rangle \Rightarrow |1\rangle$

$\Downarrow$

$X^{b_1 \oplus b_2 \oplus b_3}|0\rangle$

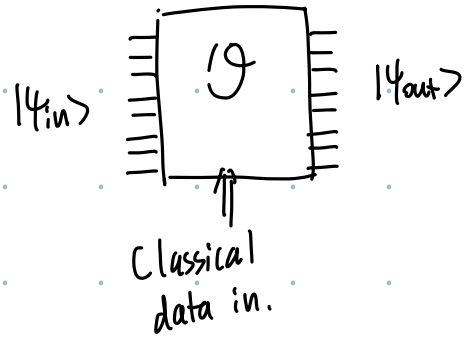
phase



$$Z \otimes Z \otimes Z |b_3 b_2 b_1\rangle = (-1)^{b_1 \oplus b_2 \oplus b_3} |b_3 b_2 b_1\rangle$$

Next big concept: the oracle.

Oracle ⇒ a black box that helps you do certain tasks.



Bottle-neck.

↳ Quantum Assisted Machine Learning

↳ QRAM

↳ Classical Info ⇒ Quantum State.

↳ Define arbitrary functions.